TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	84.7 ± 2.10
Correlation Stability	91.7 ± 0.46
Distribution Consistency	87.6 ± 1.56
Entropy Preservation	72.8 ± 5.46
Mean Z Score Drift	0.2 ± 0.04
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	7.2 ± 2.27
Data Density	100.0 ± 0.00
Coverage Score	84.5 ± 2.97
Feature Accuracy	64.8 ± 8.96
Novelty Score	0.8 ± 0.02

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Dataset Audit: thoracic_surgery.csv

1. Dataset Portrait

Attribute	Value
Rows	470
Columns	17
Numeric features	17
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
DGN	3.096	3.069	-0.027	0.9%	OK
PRE4	3.282	2.906	-0.376	11.4%	Warn
PRE5	4.569	3.305	-1.264	27.7%	Drift
PRE6	0.781	0.801	+0.020	2.5%	OK
PRE7	0.066	0.090	+0.024	37.1%	Drift
PRE8	0.145	0.170	+0.026	17.6%	Warn
PRE9	0.066	0.069	+0.003	4.8%	OK
PRE10	0.687	0.676	-0.012	1.7%	OK
PRE11	0.166	0.189	+0.023	13.8%	Warn
PRE14	11.736	11.755	+0.019	0.2%	OK
PRE17	0.074	0.066	-0.008	10.7%	Warn
PRE19	0.004	0.008	+0.004	87.5%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	84.7 ± 2.10
Correlation Stability (*2)	91.7 ± 0.46
Distribution Consistency (*3)	87.6 ± 1.56
Entropy Preservation (*4)	72.8 ± 5.46
Mean Z Score Drift (*5)	0.2 ± 0.04
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	7.2 ± 2.27

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	84.5 ± 2.97
Feature Accuracy (*10)	64.8 ± 8.96
Novelty Score (*11)	0.8 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8542553191489363	0.8521276595744682	0.8542553191489363	1.1102230246251566e-17	
F1	N/A	0.0	N/A	0	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.847872340425532	0.8521276595744682	0.8521276595744682	0.004255319148936165	
F1	N/A	0.011764705882352941	N/A	0	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	0.8553191489361703	0.8553191489361703	0	
F1	N/A	0.0	N/A	0	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.21063829787234045	0.4212765957446808	0.4053191489361702	0.19468085106382976	
F1	0.256855620661189	0.20789026882628195	0.23055157838121804	-0.026304042279970973	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8138297872340425	0.8095744680851065	0.8170212765957446	0.003191489361702116	
F1	0.11972345964178996	0.07162946384914348	0.15136083155264744	0.031637371910857456	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.820212765957446				

9

0.802127659574468

1

0.813829787234042

5

-0.00638297872340425

3

F1	0.1495587541113857	0.08428801472279733	0.14648813344465517	-0.00307062066730508	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8382978723404257	0.8393617021276597	0.8361702127659575	-0.00212765957446808	
F1	0.07535537561853352	0.03923444976076555	0.06694126620900075	-0.00841410940953274	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.847872340425532	0.8489361702127663	0.8553191489361703	0.00744680851063834	
F1	N/A	0.011764705882352941	N/A	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.851063829787234	0.8542553191489363	0.8553191489361703	0.00425531914893614	
F1	0.037675637056442016	0.011111111111111111	N/A	-0.037675637056442016	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	0.8553191489361703	0.8553191489361703	0	
F1	N/A	0.0	N/A	0	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	2.88	2.16	1.0	0.0	0.0	0.0	1.0	1.0	14.0
2	3.0	3.4	1.88	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	2.76	2.08	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	3.68	3.04	0.0	0.0	0.0	0.0	0.0	0.0	11.0
5	3.0	2.44	0.96	2.0	0.0	1.0	0.0	1.0	1.0	11.0
6	3.0	2.48	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	4.36	3.28	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	2.0	3.19	2.5	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	3.16	2.64	2.0	0.0	0.0	0.0	1.0	1.0	11.0
10	3.0	2.32	2.16	1.0	0.0	0.0	0.0	1.0	0.0	11.0
11	3.0	2.56	2.32	0.0	0.0	1.0	0.0	1.0	0.0	12.0
12	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
13	3.0	3.0	2.36	1.0	0.0	0.0	0.0	1.0	1.0	11.0
14	2.0	3.98	3.06	2.0	0.0	0.0	0.0	1.0	1.0	14.0
15	3.0	1.96	1.4	1.0	0.0	0.0	0.0	1.0	0.0	11.0
16	3.0	4.68	4.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
17	2.0	2.21	1.88	0.0	0.0	1.0	0.0	0.0	0.0	12.0
18	2.0	2.96	1.67	0.0	0.0	0.0	0.0	0.0	0.0	12.0
19	3.0	2.6	1.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
20	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.72758302357349 8	1.63809966546225 22	1.0	0.0	1.0	0.0	1.0	1.0	13.0
2	4.0	1.70379330988813 1	3.37781717242666 4	0.0	0.0	0.0	0.0	1.0	1.0	12.0
3	4.0	2.12635087971956 1	4.75668316711268 7	1.0	0.0	0.0	0.0	0.0	0.0	12.0
4	6.0	3.15876339517259 64	1.82412515700712 64	0.0	1.0	0.0	0.0	1.0	1.0	12.0
5	2.0	5.59636055506905 1	2.80432244616374 86	1.0	0.0	1.0	0.0	1.0	0.0	12.0
6	3.0	3.64032675680416 86	3.78275516080880 92	0.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	2.68	2.16	0.0	0.0	0.0	0.0	0.0	0.0	12.0
8	4.0	2.32	1.76	1.0	0.0	1.0	0.0	1.0	1.0	11.0
9	2.0	2.72	2.08	0.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.33338159212883 46	4.70473832837880 1	1.0	0.0	0.0	0.0	1.0	0.0	11.0
2	3.0	1.51156822052909 88	2.90425605399602	0.0	0.0	0.0	0.0	1.0	0.0	11.0
3	3.0	2.43624909198943 3	0.48698571070150 1	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	3.72447584652743 75	4.35679830241101 7	1.0	0.0	1.0	0.0	1.0	0.0	11.0
5	3.0	2.44352107323037 03	4.01389748261732 7	0.0	0.0	0.0	0.0	0.0	1.0	12.0
6	2.0	3.32803436506993 8	5.77826294122439 6	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	2.84	2.24	1.0	1.0	1.0	0.0	0.0	0.0	12.0
8	3.0	3.36	2.8	1.0	0.0	0.0	0.0	1.0	1.0	12.0
9	3.0	2.44	1.96	1.0	0.0	1.0	1.0	1.0	0.0	13.0
10	3.0	5.16	4.28	0.0	0.0	0.0	0.0	0.0	0.0	12.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	5.76461248306462 4	73.5064628674732 5	1.0	0.0	0.0	0.0	1.0	0.0	13.0
2	3.0	4.70205546264464 1	4.85795005242811 5	1.0	0.0	0.0	1.0	1.0	0.0	12.0
3	4.0	3.63486838251922 47	1.60937387675671 56	0.0	0.0	1.0	0.0	0.0	0.0	12.0
4	3.0	2.53905193016001 8	5.94417427065079 45	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	3.73860597912736 4	4.54750317436287 5	1.0	0.0	1.0	0.0	1.0	0.0	12.0
6	3.0	3.67050651750989 6	2.19874282407482 32	0.0	0.0	0.0	0.0	1.0	0.0	11.0
7	4.0	4.24	3.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	3.87	2.68	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	5.0	3.67	76.8	0.0	1.0	1.0	0.0	1.0	0.0	12.0
10	3.0	3.8	2.84	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	4.0	2.79633823682517 24	-0.16696238138580 677	0.0	0.0	0.0	0.0	1.0	1.0	11.0
2	3.0	1.96823246221198 07	0.35756213376976 7	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	2.44323762548463 7	1.77766014419949	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	3.09845162795695 5	-0.02653574585939 4666	1.0	0.0	0.0	0.0	1.0	1.0	12.0
5	3.0	3.46278489116791 1	2.64352736572346 7	1.0	0.0	0.0	1.0	1.0	0.0	12.0
6	5.0	5.4053962688708	0.04529551598732 917	0.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	1.84	1.28	1.0	0.0	0.0	0.0	1.0	1.0	11.0
8	5.0	4.95	4.12	1.0	0.0	0.0	0.0	0.0	1.0	11.0
9	3.0	2.92	2.28	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	1.60416497325256 2	-0.26505488949342 44	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	1.72594295429781 93	9.61516782723070 7	1.0	0.0	0.0	0.0	1.0	1.0	12.0
3	2.0	2.88811839937100 96	4.60634217996673 2	1.0	0.0	0.0	0.0	0.0	0.0	11.0
4	3.0	2.62388221590071 02	-0.16157025885462 328	1.0	1.0	0.0	0.0	1.0	0.0	11.0
5	3.0	1.86310835686106 96	1.55697901657536 16	1.0	0.0	0.0	0.0	0.0	0.0	12.0
6	3.0	2.12589206840085 55	4.63177366662269 5	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	3.0	2.44	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	2.4	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	4.36	3.92	1.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	3.4	2.8	1.0	1.0	0.0	0.0	1.0	1.0	11.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.63923499368405 65	0.36376304990323 67	0.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	3.63458989216431	1.37707900647120 1	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	4.0	2.65113044204645 5	3.42470248441426 8	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	1.70455871666933 3	4.02942926307876 8	2.0	1.0	0.0	0.0	0.0	0.0	13.0
5	2.0	2.07873952842404 8	1.85989391992324 87	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	1.13345553092924 96	4.38533144832760 5	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	2.0	3.24	2.52	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.4	1.64	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	3.24	52.3	0.0	0.0	0.0	0.0	0.0	0.0	12.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.9509634570901 36	4.99859913374390 35	2.0	0.0	0.0	0.0	0.0	0.0	11.0
2	3.0	2.6518353503612 52	5.3756567325115 48	0.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	4.10869580102574	3.8811834040654 016	1.0	0.0	0.0	0.0	1.0	1.0	12.0
4	3.0	5.8545495367260 79	3.9934415076493 65	1.0	0.0	0.0	0.0	1.0	0.0	11.0
5	3.0	4.1463649681864 22	2.3301310749327 886	0.0	0.0	0.0	0.0	1.0	1.0	12.0
6	2.0	3.1076468554683 707	4.8719010537320 05	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	2.84	2.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	2.2	1.44	1.0	0.0	0.0	0.0	1.0	0.0	12.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	7.0172410592364 72	0.3864954513605 596	1.0	0.0	0.0	0.0	0.0	0.0	12.0
2	3.0	6.8588810192840 79	-0.4913201735832 3754	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	2.0	3.6165796785052 313	1.8368280557376 724	1.0	0.0	0.0	0.0	0.0	0.0	11.0
4	3.0	4.0423697542362 77	0.1018510881987 3134	1.0	0.0	0.0	0.0	1.0	0.0	11.0
5	2.0	4.7328809595123 83	1.4364956078508 007	1.0	0.0	0.0	0.0	1.0	0.0	11.0
6	3.0	6.6522107702976 69	3.5673019472994 97	0.0	0.0	0.0	0.0	0.0	0.0	12.0
7	3.0	2.76	2.28	0.0	0.0	0.0	0.0	0.0	0.0	11.0
8	4.0	5.03	79.3	1.0	0.0	0.0	1.0	0.0	0.0	11.0
9	2.0	2.16	1.56	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.0	1.0	1.0	0.0	1.0	0.0	1.0	1.0	11.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	5.0	5.76696909159922 8	-0.21533830283759 237	2.0	0.0	0.0	0.0	0.0	0.0	14.0
2	4.0	2.26642861844559 64	0.60016159224179 82	1.0	0.0	1.0	0.0	0.0	0.0	12.0
3	3.0	4.98522068365374 25	2.79536563154255 9	1.0	0.0	0.0	1.0	1.0	0.0	12.0
4	3.0	2.69426238045316 07	0.50266503623118 06	0.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	6.23987995016903	-0.38831795138942 44	1.0	1.0	0.0	0.0	1.0	0.0	12.0
6	3.0	3.72132200088410 46	0.28304946812984 92	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	3.44	3.13	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	3.64	3.12	1.0	0.0	0.0	0.0	1.0	0.0	12.0
9	3.0	2.56	60.9	0.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	2.8	3.36	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	3.90664814866923 66	5.45283195236969 8	0.0	0.0	0.0	0.0	1.0	1.0	12.0
2	3.0	3.17496736461311 6	4.33821287814391 9	2.0	0.0	0.0	0.0	1.0	0.0	13.0
3	3.0	6.21176678903452 2	6.10472357267627 2	1.0	1.0	0.0	0.0	1.0	0.0	11.0
4	3.0	3.37960592372721 14	1.43187873372593 38	0.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	3.02152531481022 2	4.77453518657430 3	0.0	0.0	0.0	0.0	0.0	0.0	11.0
6	3.0	3.06845225573232 5	1.15333420866812 26	2.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	3.48	2.52	1.0	0.0	0.0	0.0	1.0	0.0	14.0
8	3.0	4.48	3.48	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	3.0	2.63	67.3	1.0	0.0	0.0	1.0	1.0	0.0	11.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0